

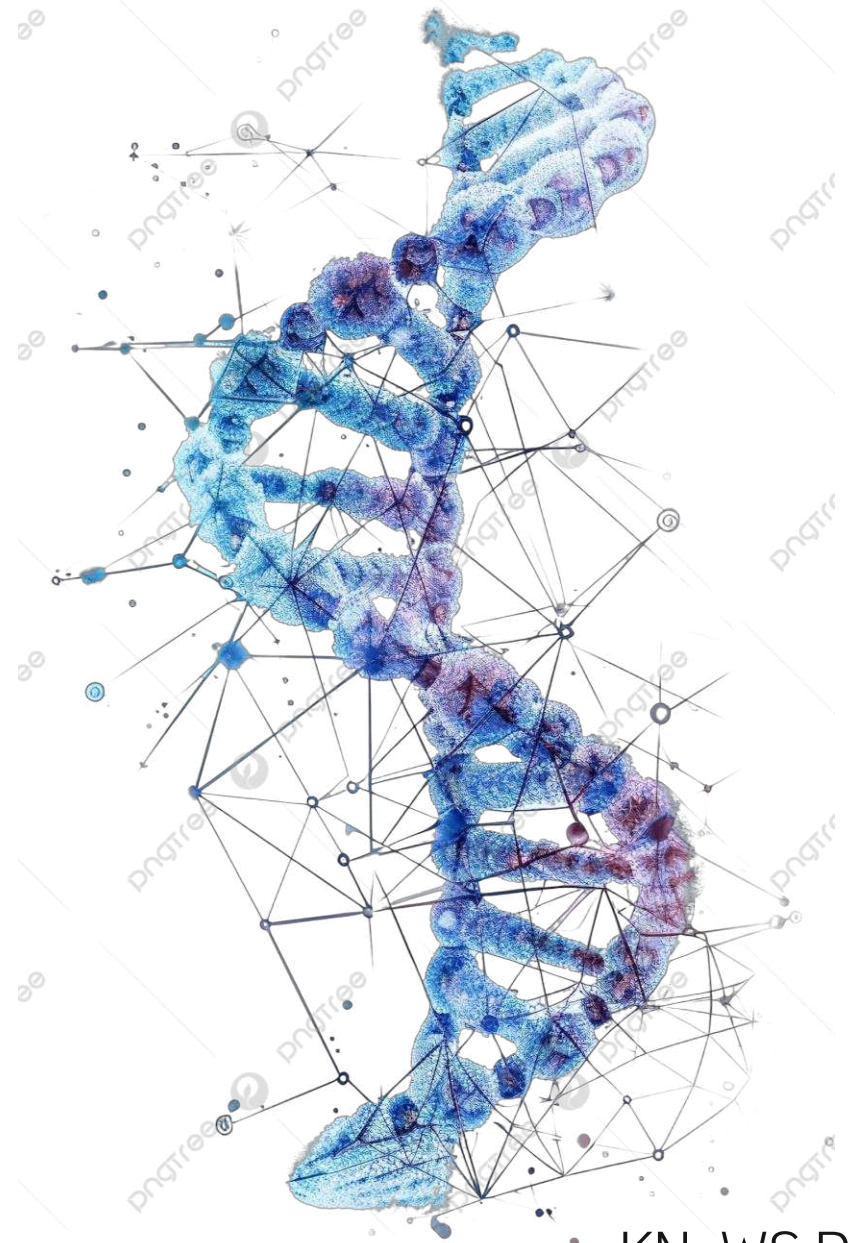
Genomic Data, Healthcare, and the Semantic Web

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KNoWS Day
March 31st, 2026

Outline



Genomic data and Healthcare?



Why integrate semantics?



How can we “translate” to semantic representations?



What can we do with semantic genomic data?

Research



Evidence base

Policy and Governance



VS.

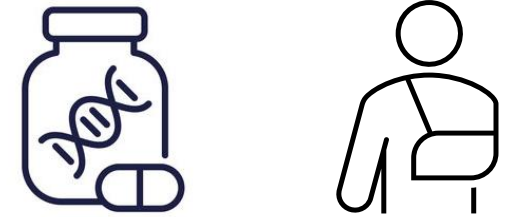


Government organizations

Genomic Medicine



Direct Clinical Care



Pharmacogenomics

Third Party Opportunities

GeneDx



Disease Screening

Barriers to Genomic Medicine?



Why do we not see more genomic data usage?

Genomic Sequence Data Sharing for Clinical Practice: A Scoping Review



Aim: Identify existing implementations, guidelines, and challenges of clinical sharing of genomic data.

Methods: Reviewed 55 studies, 57 national genomics projects, and 20 major genomics companies.

Results: Clinical genome sharing implementations exist, yet scalability and widespread adoption remain limited.

Suggested steps forward:



Can semantics help?



How can semantics help?



From a file-based approach



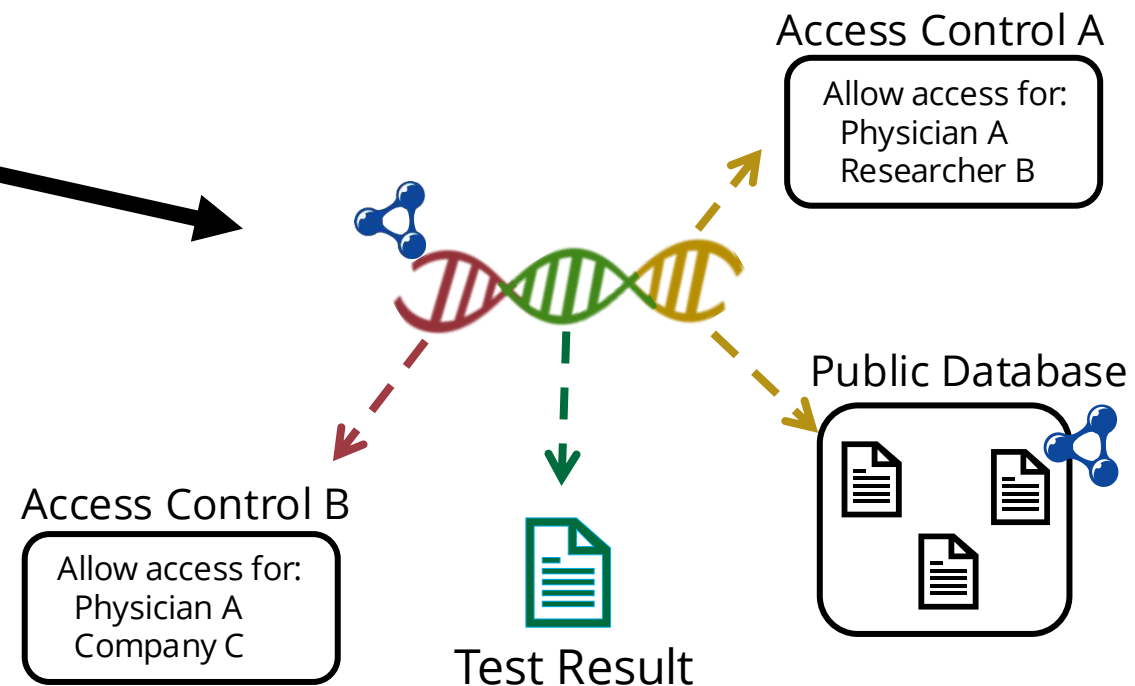
(genome data)

To a granular, dynamic RDF approach

Conversion using semantic vocabulary

Policy definition via ODRL¹

Implementation within decentralized env.



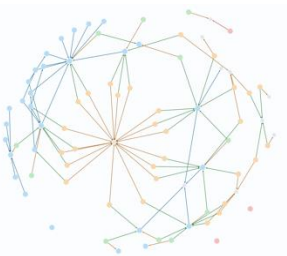
¹ <https://www.w3.org/TR/odrl-model/>

How do we get genomic data into an RDF representation?



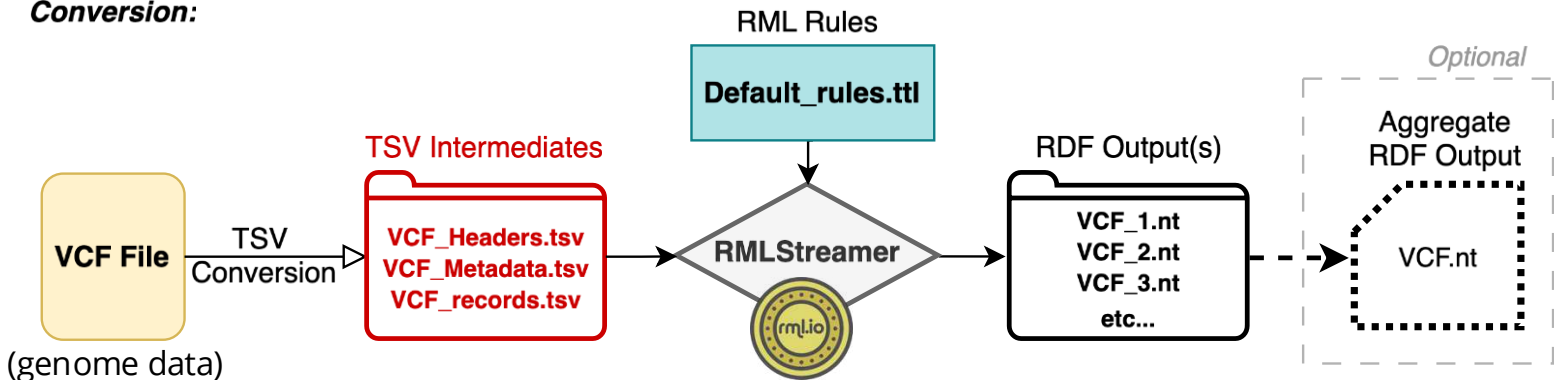
Conversion Tool

&

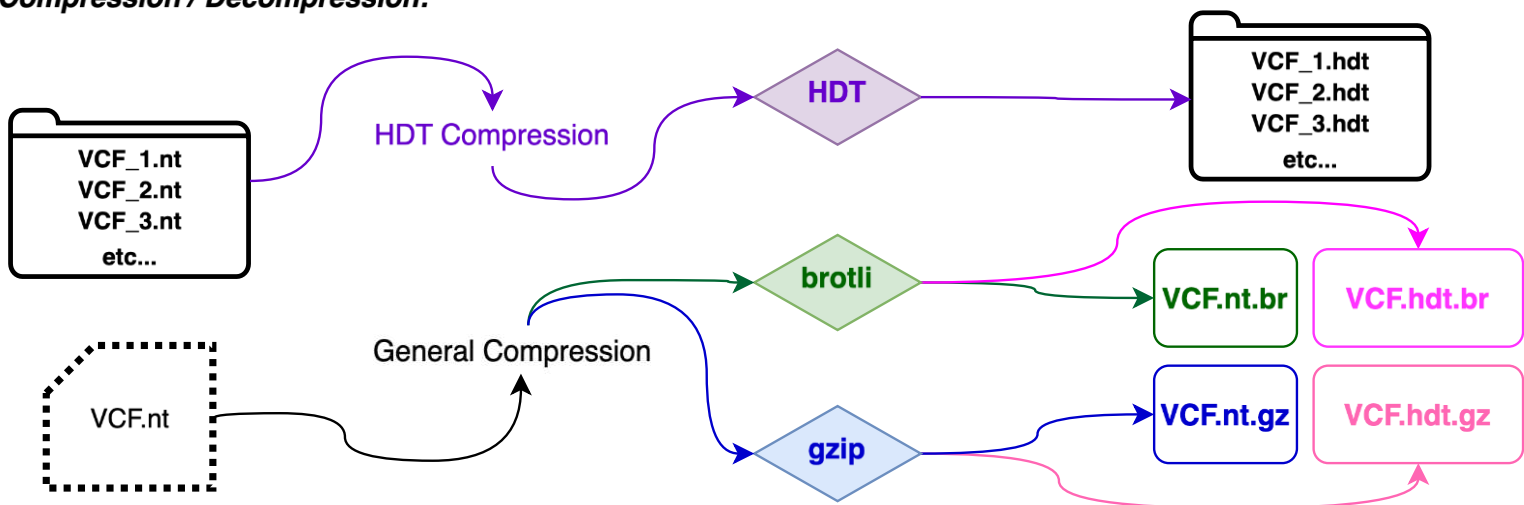


Semantic Vocabulary

Conversion:



Compression / Decompression:



* All steps are reversible

** Read [this paper](#) for more info (submitted to ECCB 2026, currently under review)

What can a genomic RDF representation enable?



1. Medically-relevant semantic data linkages:



Test Result

/



Public Data Base

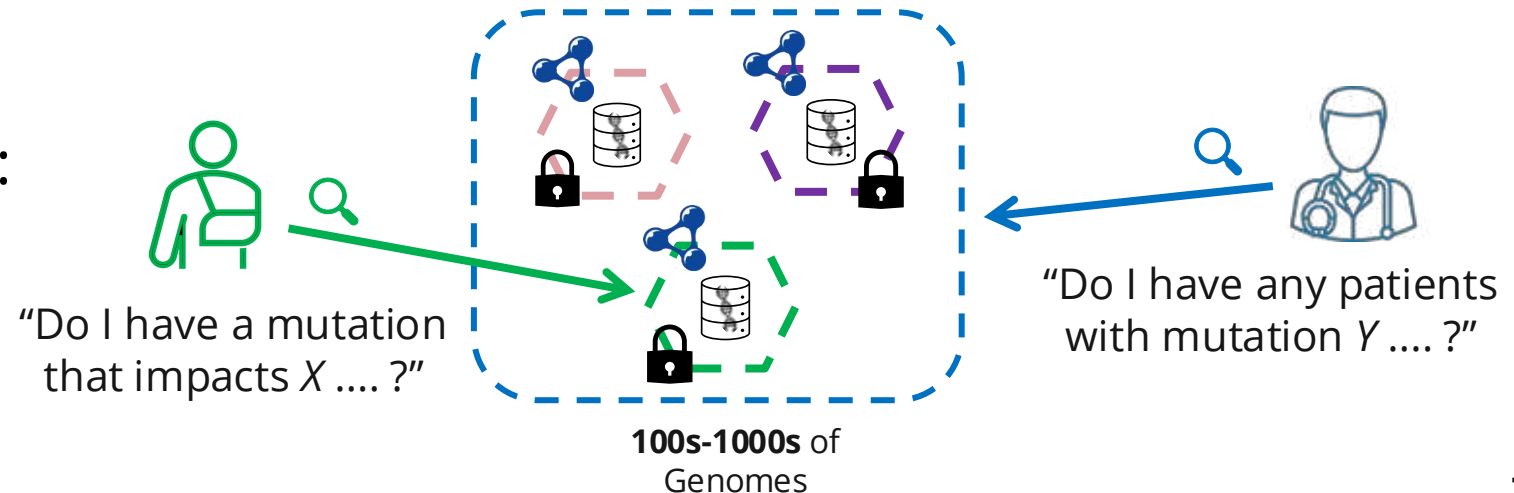
2. Granular access controls + policy definitions:
(enables cool 3rd-party healthcare + research application use-cases)



&

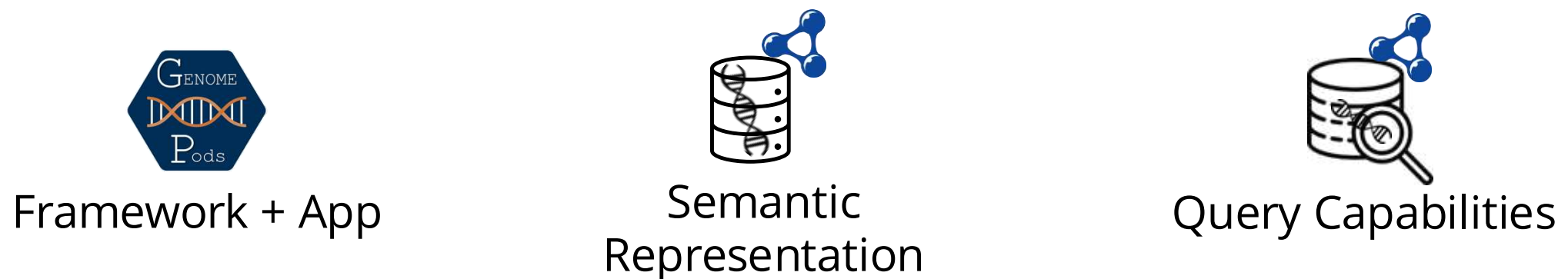
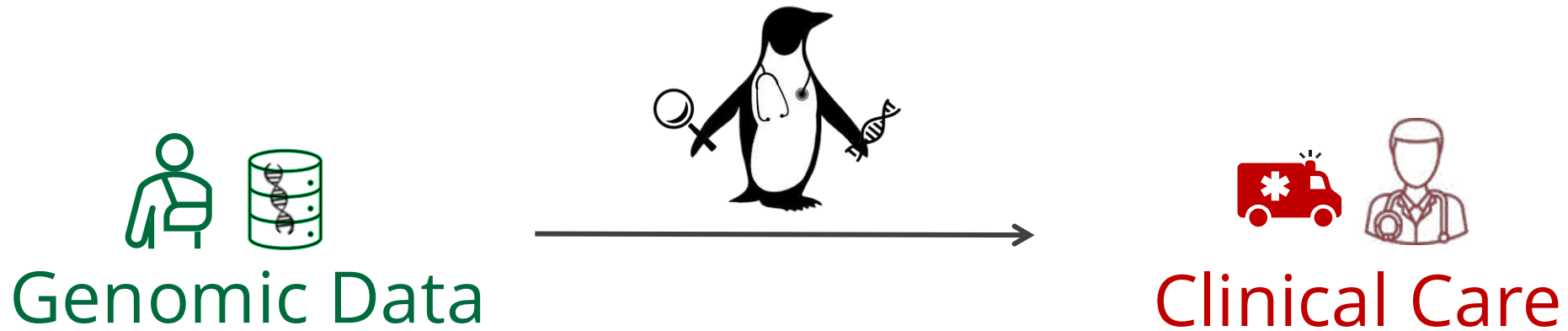


3. Federated querying capabilities:



PENGQUIN:

PErsoNal Genome QUery IN
healthcare and clinical practice



PhD Timeline

Finished:



Problem
Identification

> Comput Biol Med. 2026 Jan 1;200:111335. doi: 10.1016/j.combiomed.2025.111335.
Epub 2025 Dec 1.

**Genomic sequence data sharing for clinical practice:
A scoping review**

Elias Crum ¹, Ruben Taelman ², Bart Buelens ³, Gökhan Ertaylan ³

Affiliations + expand

PMID: 41330066 DOI: 10.1016/j.combiomed.2025.111335

Free article

FULL TEXT LINKS

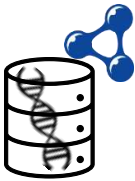
ELSEVIER
OPEN ACCESS

ACTIONS

Cite

Collections

Permalink

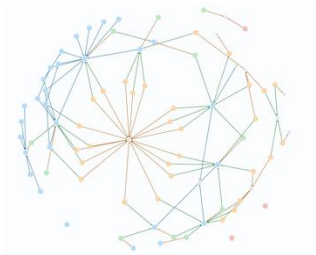


Semantic
Representation



VCF-RDFizer

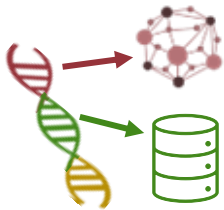
&



Future Work:



Policy
Enforcement



Data Linking

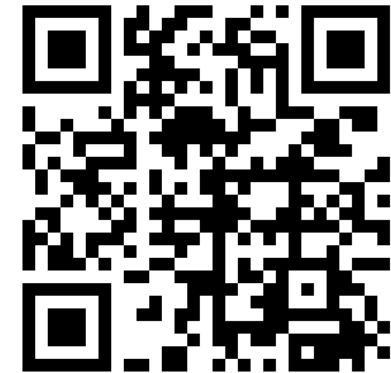


Query Capabilities

“We used to think our fate was in our stars. Now we know, in large measure, our fate is in our genes.” – James Watson



More PhD specifics



Me

Supplemental Slides

Data Interaction Framework

A Web Application to:

 *Upload Data to a Solid Pod*

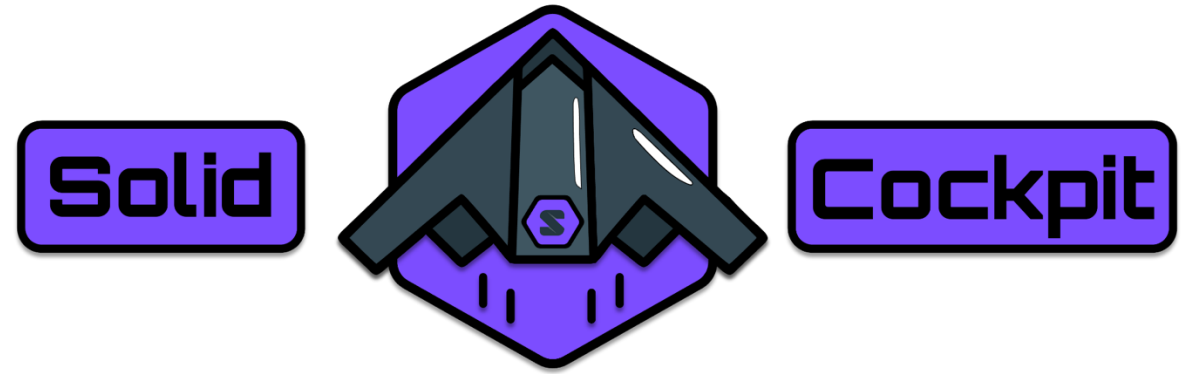
 *Browse contents of a Solid Pod*

 *Edit Privacy of (and share) data in a Solid Pod*

 *Query Data in Solid Pods + SPARQL Endpoints*

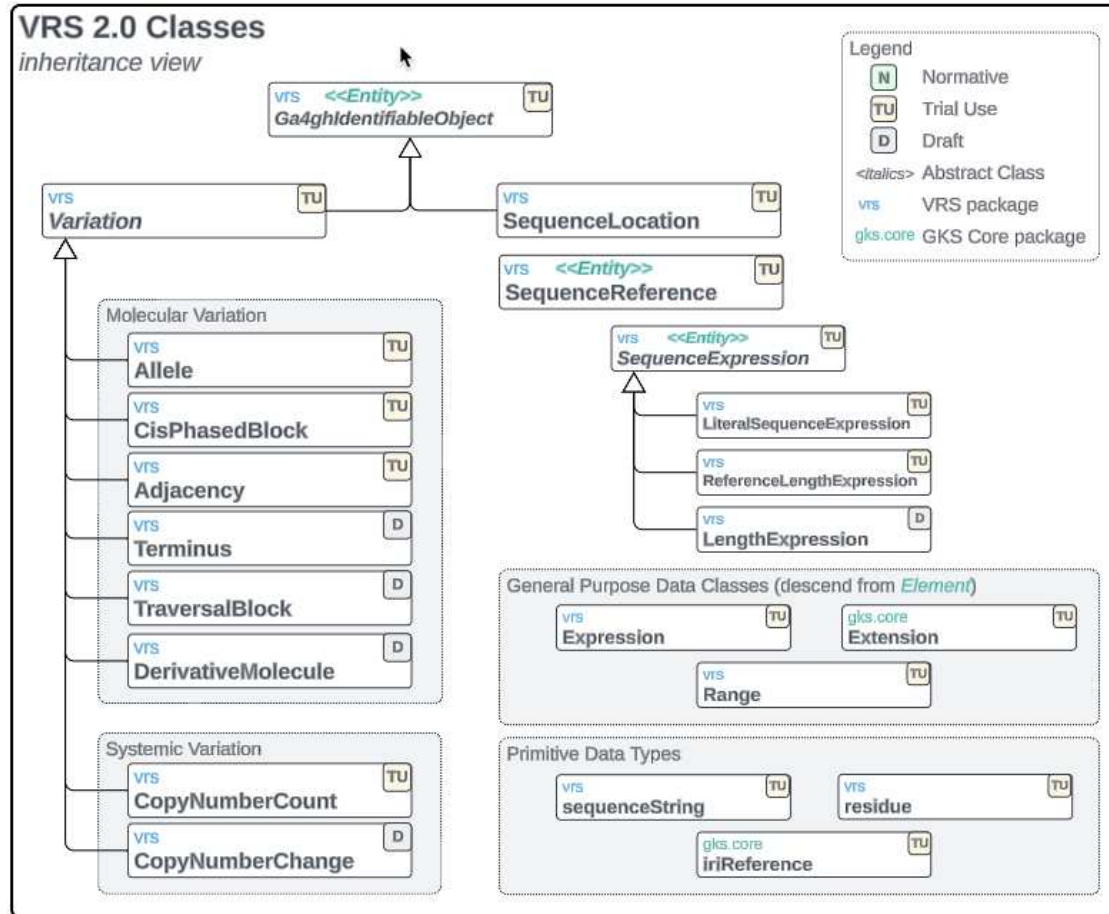
 *Includes a query cache for storing queries and results*

*** Enables patient, provider, and auditor control over data ***

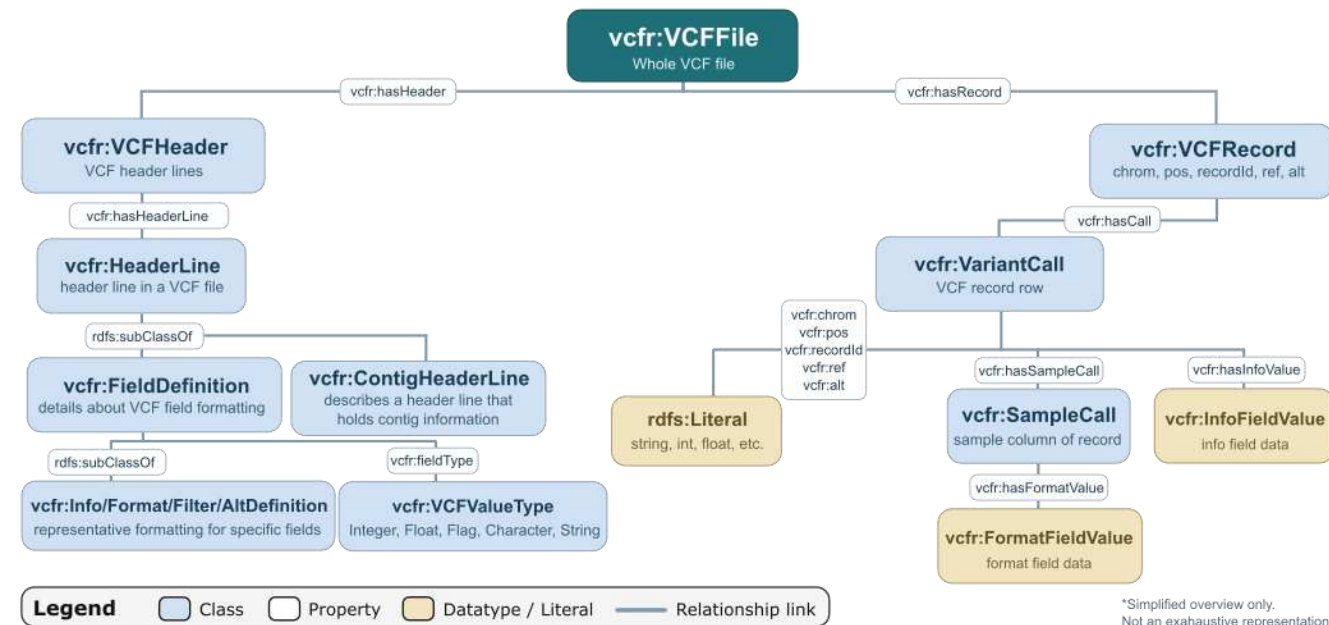


Representing VCF Semantically?

GA4GH Variant Schema:

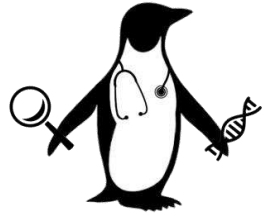


Representative VCF-RDFizer Vocabulary Class Hierarchy



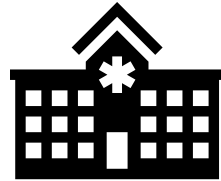
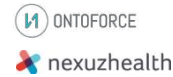
*Simplified overview only.
Not an exhaustive representation.

Envisioned Product Evolution



PhD End

PhD
Deliverable



Short Term

Start-up / Product
development
+
Hospital implementation



Medical record



Medium Term

Integrate with EHR
+
Increase application
domains



Long Term

Increase 3rd party
application support
+
Scale to Europe and
beyond